Machine Learning

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Outline

- Association Rule Mining
- Bayes Decision Theory

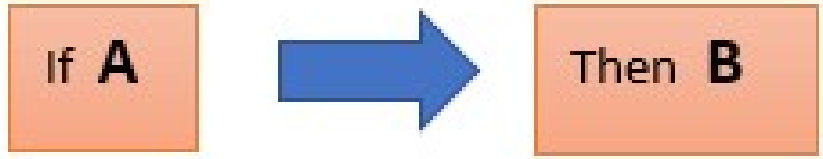
Association in Machine Learning

- In machine learning, "association" typically refers to discovering relationships or patterns between variables or features in a dataset. This concept is most commonly applied in the context of association rule learning, which is used primarily in market basket analysis and similar domains.

- The goal is to identify rules that describe how the presence or absence of one item or feature is associated with the presence or absence of another item or feature.

Key Concepts of Association

- Association rule learning works on the concept of If and Else Statement, such as if A then B
- If element is called **antecedent**, and then statement is called as **Consequent**.
- These types of relationships where we can find out some association or relation between two items is known as single cardinality.
- It is all about creating rules, and if the number of items increases, then cardinality also increases accordingly.



Key Concepts of Association

- **Itemsets:** These are collections of items or features that appear together in transactions or datasets.

- **Support:** This measures how frequently an itemset appears in the dataset. It's the proportion of transactions that include the itemset.
- Support is the frequency of A or how frequently an item appears in the dataset. It is defined as the fraction of the transaction T that contains the itemset X. If there are X itemsets, then for transactions T, it can be written as:

$$\text{Supp}(X) = \frac{Freq(X)}{T}$$

Key Concepts of Association

- **Confidence:** This measures the likelihood that an item B is present given that item A is present. It's a measure of the reliability of the rule.
- Confidence indicates how often the rule has been found to be true. Or how often the items X and Y occur together in the dataset when the occurrence of X is already given. It is the ratio of the transaction that contains X and Y to the number of records that contain X.

$$\text{Confidence} = \frac{Freq(X,Y)}{Freq(X)}$$

Key Concepts of Association

- **Lift:** This measures the strength of the association rule, indicating whether the presence of item A actually increases the likelihood of item B occurring compared to the chance occurrence of item B alone.
- It is the strength of any rule, which can be defined as below formula:

$$\text{Lift} = \frac{Supp(X,Y)}{Supp(X) \times Supp(Y)}$$

Key Concepts of Association

$$Rule\ X \Longrightarrow Y$$

$$Support = \frac{Frequency\ (X,Y)}{N}$$

$$Confidence = \frac{Frequency\ (X,Y)}{Frequency(X)}$$

$$Lift = \frac{Support}{Support(X) * Support(Y)}$$

Applications of Association

- **Market Basket Analysis:** Understanding consumer purchase behavior.

- **Recommendation Systems:** Suggesting items based on past purchase patterns.

- **Web Analytics:** Identifying patterns in user behavior on websites.

Apriori Algorithm

- Apriori is an algorithm for finding frequent itemsets and generating association rules.

- It is designed to work on the databases that contain transactions. This algorithm uses a breadth-first search and Hash Tree to calculate the itemset efficiently.

- It is mainly used for market basket analysis and helps to understand the products that can be bought together. It can also be used in the healthcare field to find drug reactions for patients.

Apriori Algorithm: Working

- **Generate Candidate Itemsets**: Create itemsets of increasing size.
- **Prune Candidates:** Remove itemsets that do not meet the minimum support threshold.
- **Generate Rules:** Create rules from the frequent itemsets that meet the confidence threshold.

Apriori Algorithm: Example

- Dataset:
- Transactions (T):
  1. {Bread, Milk}
  2. {Bread, Diaper, Beer, Eggs}
  3. {Milk, Diaper, Beer, Cola}
  4. {Bread, Milk, Diaper, Beer}
  5. {Bread, Milk, Diaper, Cola}

- Objective:

  Find frequent itemsets with a minimum support of 60%.

Apriori Algorithm: Example

- Calculate Support for 1-itemsets:**
- {Bread}: 4/5 = 80%
- {Milk}: 4/5 = 80%
- {Diaper}: 4/5 = 80%
- {Beer}: 3/5 = 60%
- {Eggs}: 1/5 = 20%
- {Cola}: 2/5 = 40%

- Frequent 1-itemsets: {Bread}, {Milk}, {Diaper}, {Beer}

Apriori Algorithm: Example

- Generate Candidate 2-itemsets from Frequent 1-itemsets:

- Candidates: {Bread, Milk}, {Bread, Diaper}, {Bread, Beer}, {Milk, Diaper}, {Milk, Beer}, {Diaper, Beer}
- Calculate Support:
- {Bread, Milk}: 3/5 = 60%
- {Bread, Diaper}: 3/5 = 60%
- {Bread, Beer}: 2/5 = 40%
- {Milk, Diaper}: 3/5 = 60%
- {Milk, Beer}: 2/5 = 40%
- {Diaper, Beer}: 3/5 = 60%
- Frequent 2-itemsets: {Bread, Milk}, {Bread, Diaper}, {Milk, Diaper}, {Diaper, Beer}

Apriori Algorithm: Example

- Generate Candidate 3-itemsets from Frequent 2-itemsets:

- Candidates: {Bread, Milk, Diaper}, {Bread, Diaper, Beer}, {Milk, Diaper, Beer}
- Calculate Support:
- {Bread, Milk, Diaper}: 2/5 = 40%
- {Bread, Diaper, Beer}: 2/5 = 40%
- {Milk, Diaper, Beer}: 2/5 = 40%

Frequent 3-itemsets: None (support threshold not met)

Apriori Algorithm: Exercise

- Transactions (T):
  1. {A, B, C}
  2. {A, B}
  3. {A, C}
  4. {B, C}
  5. {A, B, C, D}

- Objective:

  Find frequent itemsets with a minimum support of 40%.

ECLAT Algorithm

- Equivalence Class Clustering and bottom-up Lattice Traversal (ECLAT) algorithm.
- We can say that the ECLAT algorithm is an efficient and scalable version of the apriori algorithm with the following improvements.
  - The apriori algorithm works with the horizontal transaction dataset. On the contrary, the ECLAT algorithm works on a vertical data format.
  - ECLAT algorithm uses a depth-first search approach to traverse the itemsets. The Apriori algorithm uses a breadth-first approach to traverse the transaction dataset.
- The ECLAT algorithm is generally faster than the Apriori algorithm on dense datasets with a small number of distinct items and a large number of transactions.

ECLAT Algorithm: Working

- Data Transformation: Convert dataset into vertical format.
- Frequent Itemset Generation: Use a depth-first search strategy to find frequent itemsets.
- Itemset Counting: Count support by intersecting TID lists.

ECLAT Algorithm: Example

- Transactions (T):
  1. {A, B, C}
  2. {A, B}
  3. {A, C}
  4. {B, C}
  5. {A, B, C, D}

- Objective:

  Find frequent itemsets with a minimum support of 60%.

ECLAT Algorithm: Example

- Vertical Format Representation:
- {A}: {1, 2, 3, 5}
- {B}: {1, 2, 4, 5}
- {C}: {1, 3, 4, 5}
- {D}: {5}

- TID Lists:
- A: [1, 2, 3, 5]
- B: [1, 2, 4, 5]
- C: [1, 3, 4, 5]
- D: [5]

ECLAT Algorithm: Example

- Calculate Support for 1-itemsets:
- {A}: 4/5 = 80%
- {B}: 4/5 = 80%
- {C}: 4/5 = 80%
- {D}: 1/5 = 20%

Frequent 1-itemsets (≥60%): {A}, {B}, {C}

ECLAT Algorithm: Example

- Generate and Count 2-itemsets
- Candidates: {A, B}, {A, C}, {B, C}

TID List Intersection:

- {A, B}: Intersection of [1, 2, 3, 5] and [1, 2, 4, 5] → {1, 2, 5}
- {A, C}: Intersection of [1, 2, 3, 5] and [1, 3, 4, 5] → {1, 3, 5}
- {B, C}: Intersection of [1, 2, 4, 5] and [1, 3, 4, 5] → {1, 4, 5}

Support:

- {A, B}: 3/5 = 60%
- {A, C}: 3/5 = 60%
- {B, C}: 3/5 = 60%

Frequent 2-itemsets: {A, B}, {A, C}, {B, C}

ECLAT Algorithm: Example

- Generate and Count 3-itemsets

Candidates: {A, B, C}

- TID List Intersection:
- {A, B, C}: Intersection of {1, 2, 5}, {1, 3, 5}, {1, 4, 5} → {1, 5}
- Support:
- {A, B, C}: 2/5 = 40%

Frequent 3-itemsets: None (support threshold not met)

ECLAT Algorithm: Exercise

- Transactions (T):
  1. {A, B, E}
  2. {B, D}
  3. {B, C}
  4. {A, B, D}
  5. {A, C}
  6. {A, B, C}

- Objective:

  Find frequent itemsets with a minimum support of 60%.

Basic Statistics Definitions

Definitions

- A **random experiment** is a physical situation whose outcome cannot be predicted until it is observed.
- A **sample space** is a set of all possible outcomes of a random experiment.
- A **random variable** is a variable whose possible values are numerical outcomes of a random experiment.
  - **Discrete Random Variable** is one which may take on only a countable number of distinct values.
  - **Continuous Random Variable** is one which takes an infinite number of possible values

Definitions

- **Probability** is the measure of the likelihood that an event will occur in a Random Experiment.

- **Conditional Probability** is a measure of the probability of an event given that (by assumption, presumption, assertion or evidence) another event has already occurred.

Conditional Probability Formula

$$\underbrace{P(A \mid B)}_{\text{Probability of } A \text{ given } B} = \frac{\overbrace{P(A \cap B)}^{\text{Probability of } A \text{ and } B}}{\underbrace{P(B)}_{\text{Probability of } B}}$$

Definitions

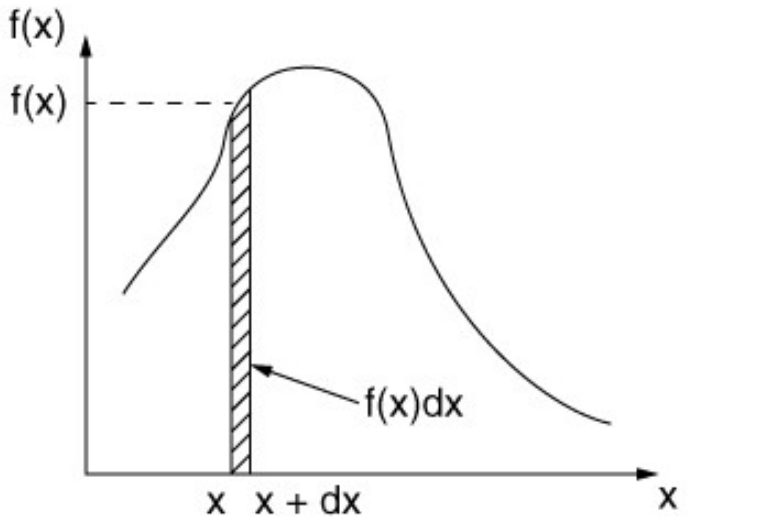
- The **probability distribution** for a random variable describes how the probabilities are distributed over the values of the random variable.
- For a discrete random variable, x, the probability distribution is defined by a **probability mass function**, denoted by f(x). This function provides **the probability for each value** of the random variable.

  - f(x) must be nonnegative for each value of the random variable
  - the sum of the probabilities for each value of the random variable must equal one.

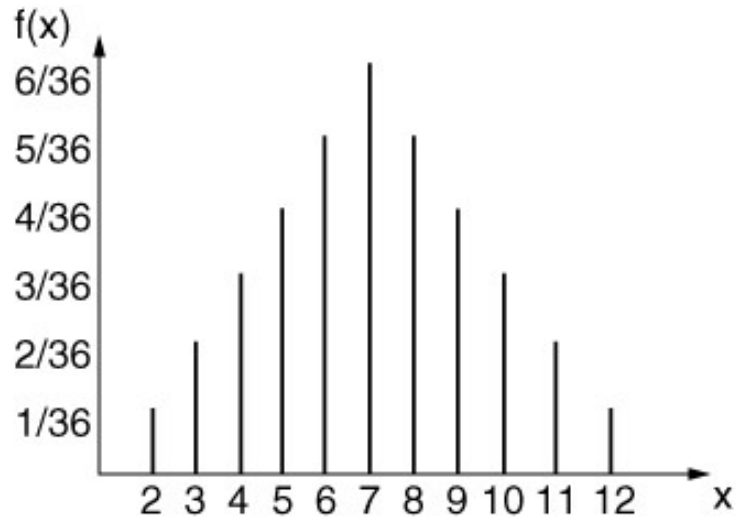
Definitions

- In the continuous case, the counterpart of the probability mass function is the **probability density function**, also denoted by f(x).
- The area under the graph of f(x) **corresponding to some interval**, obtained by computing the **integral of f(x) over that interval**, provides the probability that the variable will take on a value within that interval.
  - f(x) must be nonnegative for each value of the random variable.
  - The integral over all values of the random variable must equal one.

Definitions



Continuous probability density function



Probability mass function

Probability Density Function

F(x) = $P(a \leq x \leq b) = \int_a^b f(x)dx \geq 0$

$$p(x) = P(X = x)$$

The probability of x = the probability(X = one specific x)

Feature

- A feature is a **distinctive trait** or a special attraction.
- In machine learning, a feature is an individual measurable property or characteristic of a phenomenon.

- **Feature:** is a list of numbers eg: age, name, height, weight etc., that means every column is a feature in relational table.
- **Feature Vector** is representation of particular row in relational table. Each row is a feature vector, row 'n' is a feature vector for the 'n'th sample.
- **Feature Set:** Help to predict the output variable.

ID	First Name	Last Name	Email	Year of Birth
1	Peter	Lee	plee@university.edu	1992
2	Jonathan	Edwards	jedwards@university.edu	1994
3	Marilyn	Johnson	mjohnson@university.edu	1993
6	Joe	Kim	jkim@university.edu	1992
12	Haley	Martinez	hmartinez@university.edu	1993
14	John	Mfume	jmfume@university.edu	1991
15	David	Letty	dletty@university.edu	1995

Feature Vector

Table: Students

Bayes Theorem

- Bayes' Theorem is a simple mathematical formula used for calculating conditional probabilities.
- Conditional probability is a measure of the probability of an event occurring given that another event has (by assumption or evidence) occurred.

Probability of B occurring given evidence A has already occurred

Probability of A occurring

$$P(\mathrm{A}|\mathrm{B}) = \frac{P(\mathrm{B}|A) \cdot P(A)}{P(B)}$$

Probability of A occurring given evidence B has already occurred

Probability of B occurring

Naïve Bayes

LIKELIHOOD

The probability of "B" being True, given "A" is True

PRIOR

The probability "A" being True. This is the knowledge.

$$P(A|B) = \frac{P(B|A).P(A)}{P(B)}$$

POSTERIOR

The probability of "A" being True, given "B" is True

MARGINALIZATION

The probability "B" being True.

Definitions: Bayes Theorem

- **Priors:** How likely it is for an event to occur in nature.
  - It is also called state of nature
  - Priors are known before the training process. It is usually pre-determined and depends on the external factors.
- **Likelihood:** It is the probability of how likely a feature occurs given that it belongs to a particular class.
  - Also called class conditional probability.
  - It is denoted by, $P(X|A)$ where x is a particular feature
  - It is the quantity that we have to evaluate while training the data.
- **Evidence:** It is the probability of occurrence of a particular feature i.e. $P(X)$.
  - it is merely a scaling factor that guarantees that the posterior probabilities sum to one.

Definitions: Bayes Theorem

- **Posterior:** It is the probability of occurrence of a class when certain features are given:
  - The posterior represents the probability that an observation falls into class A or B given the measurement x.
  - The output of applying Bayes Theorem.
  - Each observation receives a posterior probability for every class, and all the posteriors must add up to 1.

Bayes Decision Theory

- Bayesian decision theory refers to the statistical approach based on tradeoff quantification among various classification decisions based on the concept of probability (Bayes Theorem) and the costs associated with the decision.

- It makes the assumption that the decision problem is posed in probabilistic terms, and that all the relevant probability values are known.

Bayes Decision Theory

- Bayesian Decision Theory is a statistical framework for making the best decision when dealing with uncertainty by using probabilities to minimize the risk of making an error.

- It works by combining prior knowledge (what you know before) with new evidence to calculate posterior probabilities, then choosing the action that maximizes expected utility (benefit) or minimizes expected loss.

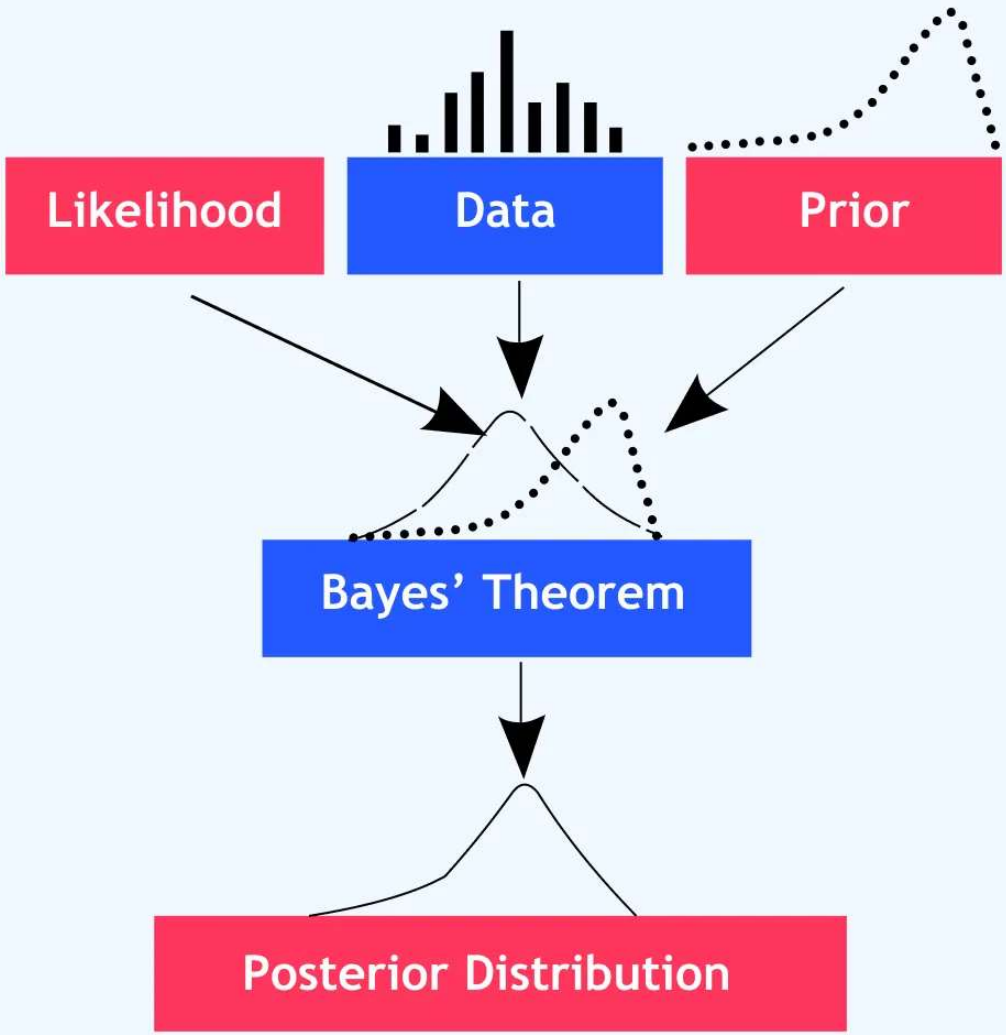
Bayes Decision Theory

- **The Problem:** You need to make a decision (e.g., classifying an image as a cat or dog) but there's uncertainty involved.
- **Prior Knowledge:** You start with some prior beliefs about the different possibilities, like the general probability of seeing a cat versus a dog.
- **New Evidence:** You gather some data or evidence (e.g., features of the image).
- **Bayes' Theorem:** You use Bayes' Theorem to update your prior knowledge with the new evidence to get a posterior probability. This is the probability of something being true, given the evidence you've observed.

Bayes Decision Theory

- **Loss/Utility:** You assign a numerical value (utility) to the potential outcomes or a cost (loss) for making the wrong decision.
- **Decision Rule:** You choose the action that has the highest posterior probability of being correct, or, more formally, the one that maximizes your total expected utility or minimizes your expected loss.

Bayes Decision Theory



Summary

- Association Rule Mining
- Bayes Decision Theory